

When to Use

Use Pearson's correlation when you have two quantitative (continuous) variables and want to test whether there is a linear relationship between them.

H Research Hypothesis: Over time, the age gap between the actors playing Lois Lane and Clark Kent has decreased, such that older portrayals will have larger age differences.

H Null Hypothesis: There is no linear relationship between year and the age difference between the actors playing Lois Lane and Clark Kent.

Running a Correlation

- 1 **Analyze → Correlate → Bivariate**
- 2 Highlight each of the desired variables and click the arrow button
- 3 Be sure "Pearson" and "Two-tailed" buttons are checked
- 4 Click "Options" — check "Means and standard deviations"
- 5 Click "OK" to run the analysis

SPSS Syntax

CORRELATIONS

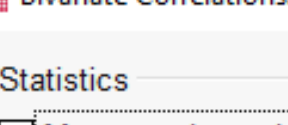
```
/VARIABLES=year age_diff
/PRINT=TWOTAIL
/STATISTICS=DESCRIPTIVES
/MISSING=LISTWISE.
```

- 1
- 2
- 3
- 4

- 1 List the variables to correlate
- 2 Get 2-tailed p-values (use ONETAIL for directional hypothesis)
- 3 Get means and standard deviations
- 4 Use listwise deletion (use PAIRWISE to retain more cases)

Note

The **standard deviation** (SD) tells about the expected variation of individual scores around the mean of the distribution.



 Bivariate Correlations: Options ✕

Statistics

☒ Means and standard deviations

☐ Cross-product deviations and covariances

Missing Values

☒ Exclude cases pairwise

☐ Exclude cases listwise

[Continue](#) [Cancel](#) [Help](#)